



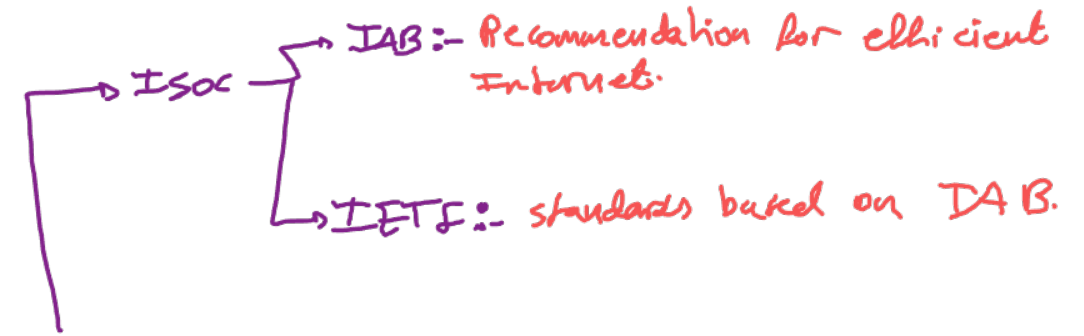
Security on
Router.

IP Security

CHAPTER 20

* App just say I want SSL & SSL
implements everything, Not the user.
Traffic on the one application-
end systems
* TCP header Not secure w/ SSL
before it.

IP SECURITY OVERVIEW



- RFC 1636. Issued in 1994 by the Internet Architecture Board (IAB)

Documents :-

- Identifies key areas for security mechanisms

- need to protect all communication between two end systems
Not Appor user.
نی ستور
از
له
- Need to secure the network infrastructure from **unauthorized monitoring and control of network traffic**
- Need to secure **end-user-to-end-user traffic** using **authentication and encryption** mechanisms

- IAB included **authentication** and **encryption** as **necessary** in **IPv6**

- The IPsec specification now exists as a set of Internet standards

Ipsec integrated with IPv6
↑ supports
built in + backward
compatibility with IPv4
- just to say

APPLICATIONS OF IPSEC

مزايا ال Ip security :-

- IPsec can provide security between any two network-layer entities

host-host, ^{gateway gateway gateway} host-router, router-router
^{بجته بجته بجته}

host:- Application layer entity (means have transport + Network + D.L)
Router:- Network layer header. [add + Read headers. Read + modify] network layer + DL layer.

- Examples include:

- Secure branch office connectivity over the Internet ^{site to site security. ex:- have branch in agaba.}
- Secure remote access over the Internet ^{دندو له ز تاي وال Traffic بين ويبين ايجاز ال secure}
- Establishing extranet and intranet connectivity with partners ^{access من بيا البتعة}
- Used widely to establish VPNs ^{بشخص كجته واسر}

- Principal feature of IPsec is that it can encrypt and/or authenticate all traffic at the IP level

- Thus all distributed applications (remote logon, client/server, e-mail, file transfer, Web access) can be secured ^{all application / all users are secure without any of them involved}

صار كاسر blanket بتحمي القوتها.

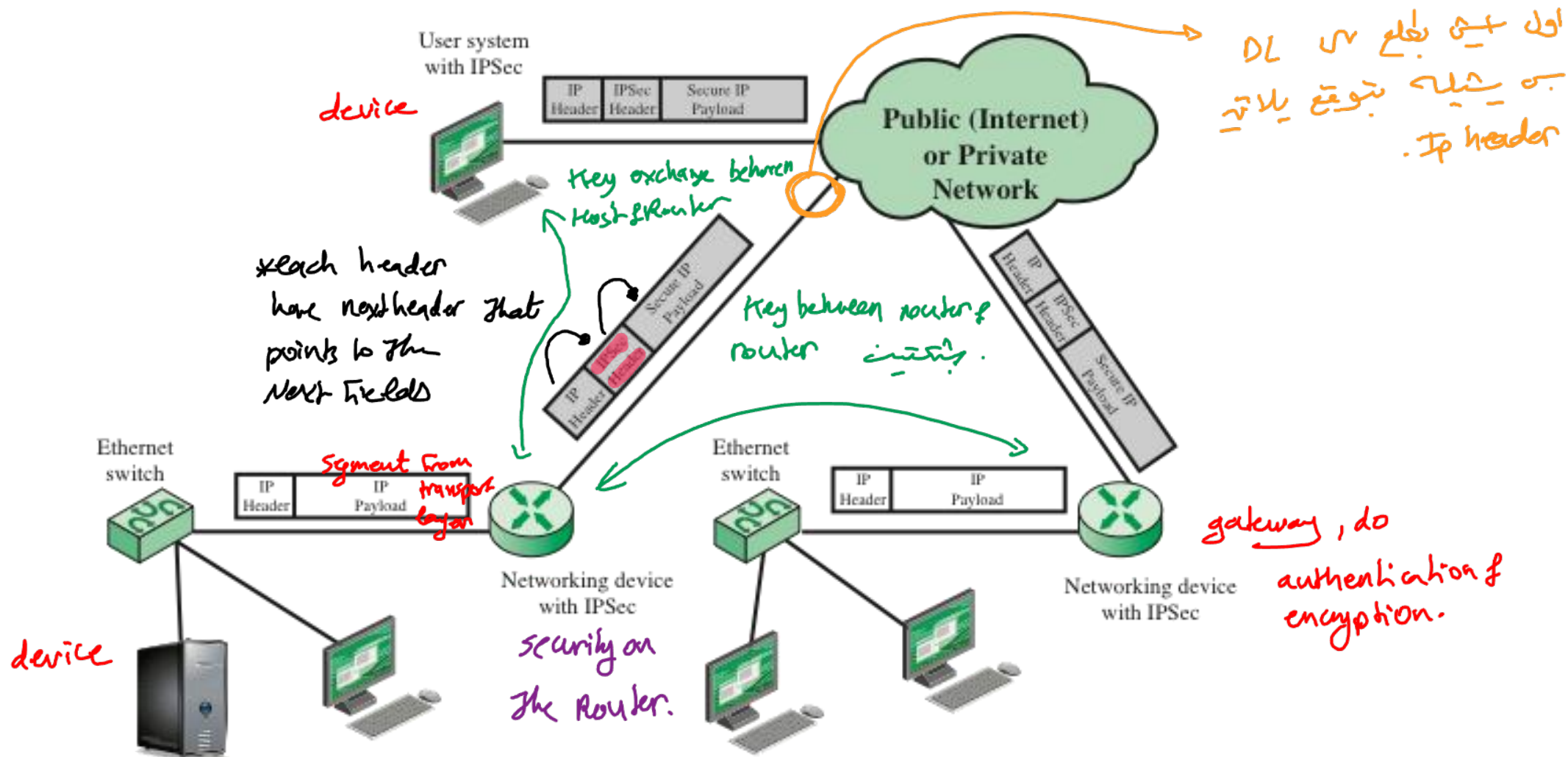


Figure 20.1 An IP Security Scenario

BENEFITS OF IPSEC

Some of the benefits of IPsec:

1. When IPsec is implemented in a **firewall or router**, it provides strong security that can be applied **to all traffic crossing the perimeter** *اهم كل ال traffic الطالعش ، او بالعنى اي traffic جا يتـ*
2. IPsec in a firewall is resistant to bypass if all traffic from the outside must use IP and the firewall is the only means of entrance from the Internet into the organization
3. IPsec is **below** the transport layer (TCP, UDP) and so is **transparent to applications** *بغير اهم كل شي بغفد النفا
ال application وة ال user . involved*
4. *machin* End-to-end or hop-by-hop security. *Router*
5. IPsec is **transparent to end users**
 - There is no need to train users on security mechanisms, issue keying material on a per-user basis, or revoke keying material when users leave the organization
6. IPsec can provide security for individual users' machines if needed *secure all the machines*
 - This is useful for offsite workers and for setting up a secure virtual subnetwork within an organization for sensitive applications

ROUTING APPLICATIONS

- IPsec can play a vital role in the ~~routing architecture~~ required for internetworking
**Secure update + advertisement for router.*

- IPsec can assure that:

1. ~~A router advertisement~~ (a new router advertises its presence) comes from an authorized router
2. A router seeking to establish or ~~maintain a neighbor relationship~~ with a router in another routing domain is an authorized router
3. ~~A routing update~~ is not forged

IPSEC SERVICES

- IPsec provides security services at the IP layer by enabling a system to: *3 services:-*

1. Select required security protocols *AH / ESP*
2. Determine the algorithm(s) to use for the service(s) *use existing ones.*
3. Put in place any cryptographic keys required to provide the requested services *options for key exchange.*

2 main protocols:-

- Two protocols are used to provide security:

1. *Authentication Header (AH):* provides authentication service only; and *authentication.*
2. *Encapsulating Security Payload (ESP):* provides combined *encryption & optional authentication.*
encryption/authentication services.

why needed if theres ESP?
① don't always need to do encryption
② it can authenticate the header.

always

TRANSPORT AND TUNNEL MODES *2 modes of operations :-*

EH / ESP

Transport Mode

- Provides protection primarily for **upper-layer protocols**
- Typically used for **end-to-end communication** between two hosts *two end systems have key.*

AH / ESP

Tunnel Mode *packet 10-21*

- Provides protection to the **entire IP packet**
- Used when one or both ends of a security association (SA) are a **security gateway** *Host-Router, Router-Host, Router-Router, Not end, to end, one of the Router involved.*
 - Therefore, a number of hosts on networks behind firewalls may engage in secure communications without implementing IPsec

- ESP in transport mode encrypts and optionally authenticates the IP payload **but not the IP header** *Just works on payload not in the header.*

- AH in transport mode authenticates the IP payload and **selected portions of the IP header** *hash (Payload + key) + TTL*
segment *These parts change along the path so when Receiver verify it will not be the same.*

- ESP in tunnel mode encrypts and optionally authenticates the entire **inner IP packet including the inner IP header** *Payload, transport header, Network layer header*
- AH in tunnel mode authenticates the **entire inner IP packet and selected portions of the outer IP header** *should be connected*

only verify IPsec header in Src → Router. des → another Router.

Table 20.1

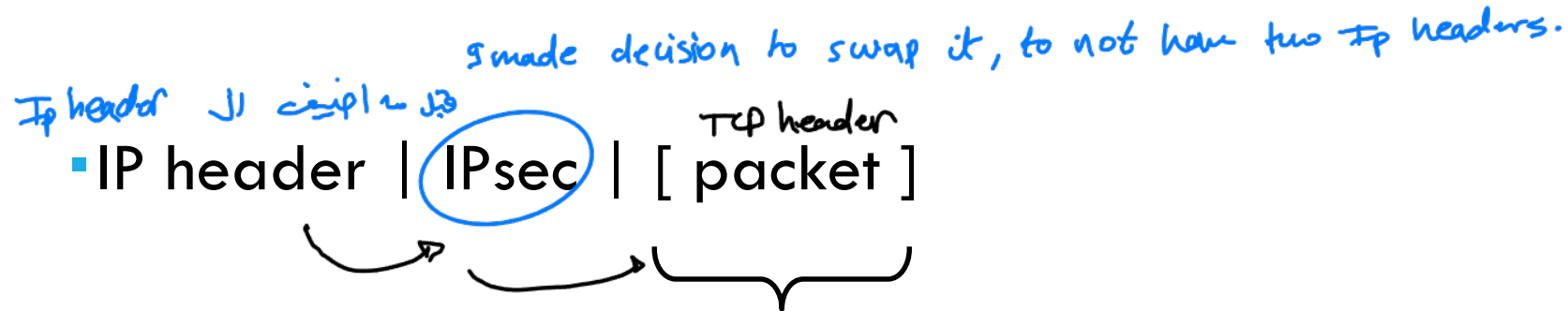
inner ip header
src → Initial src.
dst → final dst.

Tunnel Mode and Transport Mode Functionality

	Transport Mode SA	Tunnel Mode SA
AH	Authenticates IP payload and selected portions of IP header and IPv6 extension headers.	Authenticates entire inner IP packet (inner header plus IP payload) plus selected portions of outer IP header and outer IPv6 extension headers.
ESP	Encrypts IP payload and any IPv6 extension headers following the ESP header.	Encrypts entire inner IP packet.
ESP with Authentication	Encrypts IP payload and any IPv6 extension headers following the ESP header. Authenticates IP payload but not IP header.	Encrypts entire inner IP packet. Authenticates inner IP packet.

TRANSPORT VS. TUNNEL MODE

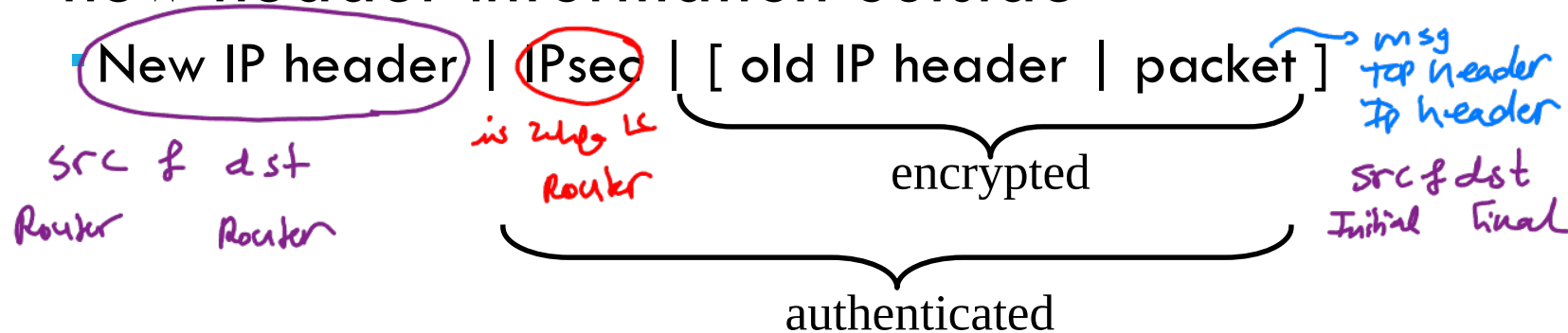
- **Transport mode:** original IP header not touched; IPsec information added between IP header and packet body



- Most logical when IPsec used end-to-end

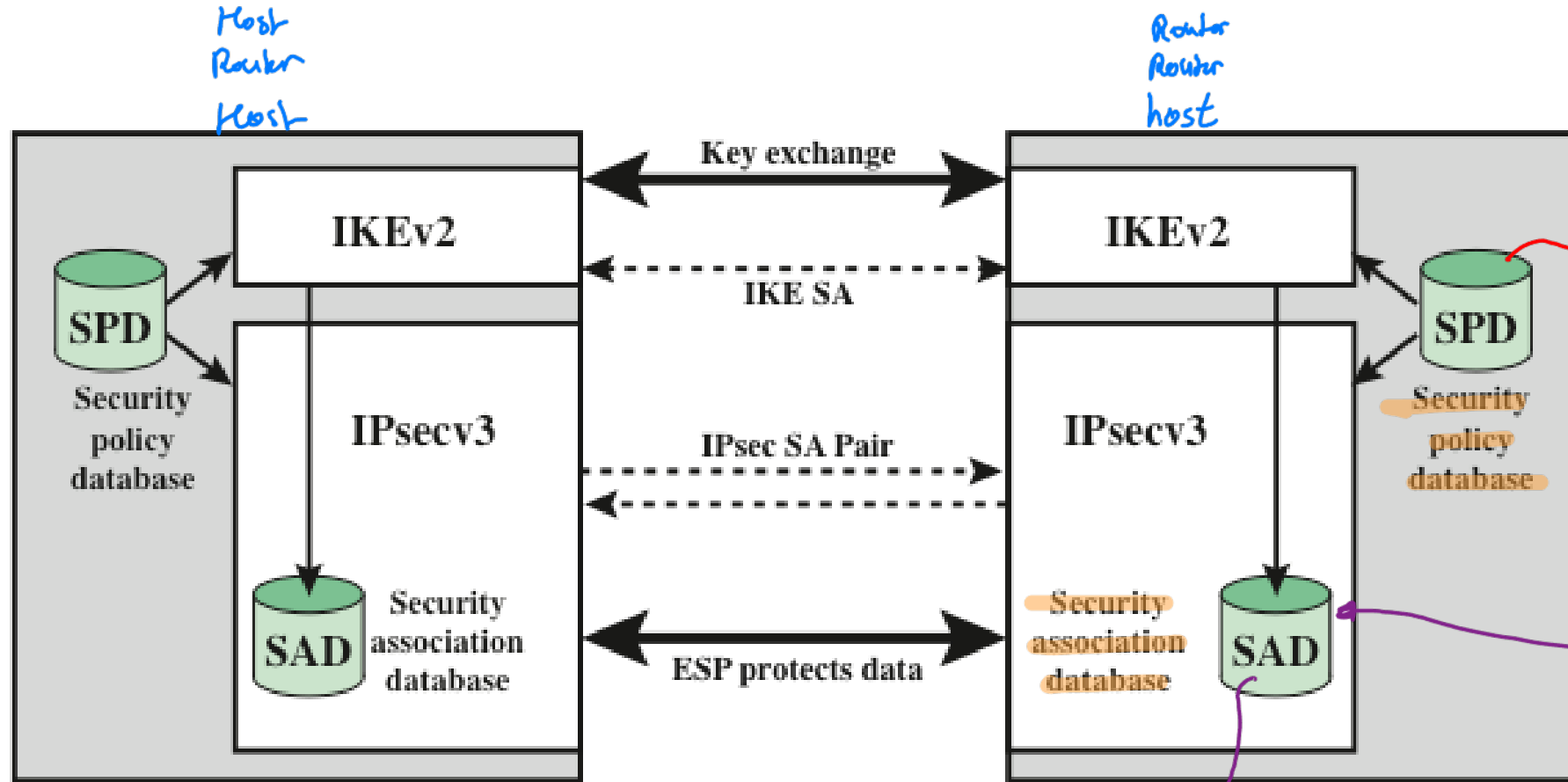
TRANSPORT VS. TUNNEL MODE

- **Tunnel mode:** keep original IP packet intact but protect it; add new header information outside



- Can be used when IPsec is applied at intermediate point along path (e.g., for firewall-to-firewall traffic)
 - Treat the link as a secure tunnel
- Results in slightly longer packet

*Two data bases of key exchange:-



معلومہ ال IP security سٹورج -
Database of secure connections.

{ sequence number / keys / authentication / encryption }

Figure 20.2 IPsec Architecture

one way secure connection
need at least two S A D

ما بینہ ال فہمے . وہی تہہ

SAD و SPI و SAD

بہرہ SAD و SPI . بینہ ال src و dst

SECURITY ASSOCIATION (SA)

- A one-way connection between a sender and a receiver that affords security services to the traffic carried on it.

- In any IP packet, the SA is uniquely identified by three parameters:

1. Security Parameters Index (SPI) رقمها

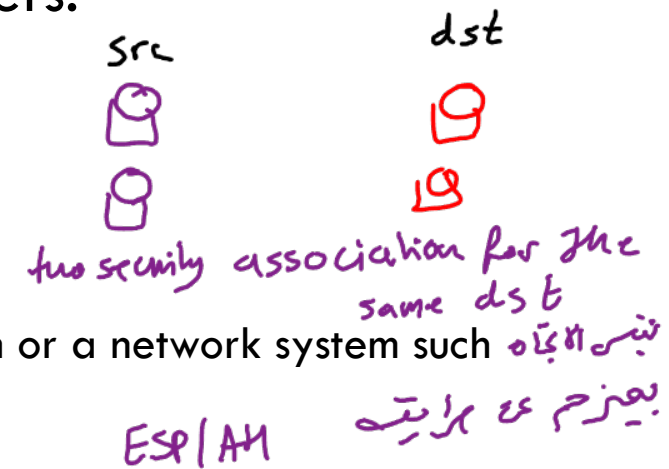
- A 32-bit unsigned integer assigned to this SA and having local significance only

2. IP Destination Address

- Address of the destination endpoint of the SA, which may be an end-user system or a network system such as a firewall or router

3. Security protocol identifier AH/ESP

- Indicates whether the association is an AH or ESP security association



SECURITY ASSOCIATION DATABASE (SAD)

- Defines the parameters associated with each SA
- Normally defined by the following parameters in a SAD entry:
 - Security parameter index (32 bit ID)
 - Sequence number counter (32 bits, anti replay)
 - Sequence counter overflow (flag bit for action on overflow)
 - Anti-replay window (to know received packet if it is replayed)
 - AH information (auth alg, keys, key lifetime, and AH parameters)
 - ESP information (Auth and Enc alg, Keys, key lifetimes, other parameters)
 - Lifetime of this security association
 - IPsec protocol mode (tunnel, transport)
 - Path MTU

1 → new security Association, 0 → overflow continue

protect from Replay together.

SECURITY POLICY DATABASE *what to do with the packet either to* *طريقة او كيفية*

- Nodes maintain a table specifying what is required for each incoming packet
 - Discard
 - Bypass without IPsec protection
 - Require IPsec protection *from SAD*
 - Auth only
 - Enc only
 - Both
- As with firewalls, decisions can be based on any information in the packet

SECURITY POLICY DATABASE (SPD)

- Each SPD entry is defined by a set of IP and upper-layer protocol field values called **selectors** دہا fields بال packet پلے اہل
عہہ امن ہے policy تنظیم
کیا۔
- These are used to filter outgoing traffic in order to map it into a particular SA
- In complex environments:
 - there may be multiple entries that potentially relate to a single SA or
 - multiple SAs associated with a single SPD entry

SPD SELECTORS

Fields to build the policies on.

• The following selectors determine an SPD entry:

• **Remote IP address** البعاز البع ماع

- single IP, an enumerated list or range of addresses, or a wildcard (mask) address
- The latter two are required to support more than one destination system sharing the same SA

• **Local IP address** my Ip addr. could I be src if received or dst if send.

- single IP, an enumerated list or range of addresses, or a wildcard (mask) address
- The latter two are required to support more than one source system sharing the same SA

• **Next layer protocol** either TCP or UDP or ICMP. {my policy to protect just ICMP, or TCP or UDP} or all

- IP protocol header includes a field that designates the protocol operating over IP

• **Name** Not real selector, there is no field Name in packet. Just saved in SPD as info only.

- A user identifier from the operating system or a name for the SPD entry
- Not a field in the IP or upper-layer headers but is available if IPsec is running on the same operating system as the user

• **Local and remote ports** secure TCP & port 80 or UDP & port 53. or all I want to secure all ports.

- These may be individual TCP or UDP port values, an enumerated list of ports, or a wildcard port

If I send → Remote is the dst Remote.
If I received → Remote is the src

Table 20.2

Host SPD Example

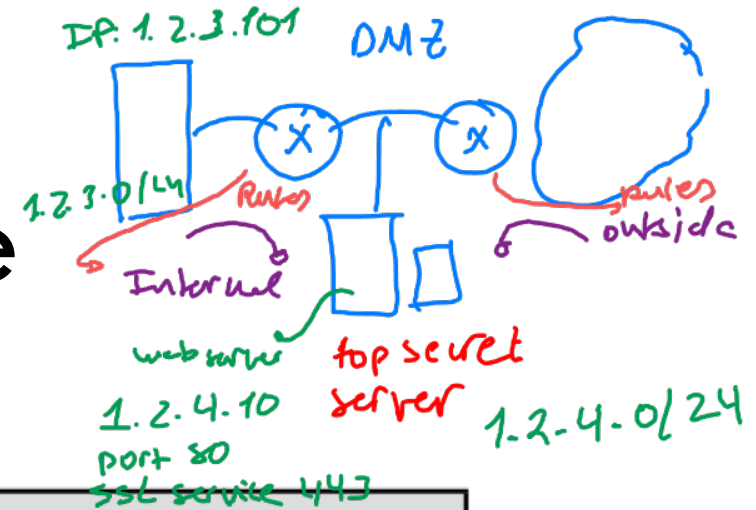


Table 20.2 Host SPD Example

Protocol	Local IP	Port	Remote IP	Port	Action	Comment
UDP	1.2.3.101	500	*	500	BYPASS	IKE Internet key exchange
ICMP	1.2.3.101	*	*	*	BYPASS	Error messages
*	1.2.3.101	*	1.2.3.0/24	*	PROTECT: ESP intransport-mode	Encrypt intranet traffic go to S.A.D.
TCP	1.2.3.101	*	1.2.4.10	80	PROTECT: ESP intransport-mode	Encrypt to server
TCP	1.2.3.101	*	1.2.4.10	443	BYPASS	TLS: avoid double encryption
*	1.2.3.101	*	1.2.4.0/24	*	DISCARD	Others in DMZ
*	1.2.3.101	*	*	*	BYPASS	Internet

any packet

بypass
بypass

anyone
in DMZ

ممنوع ايجي من ال
DMZ
web server

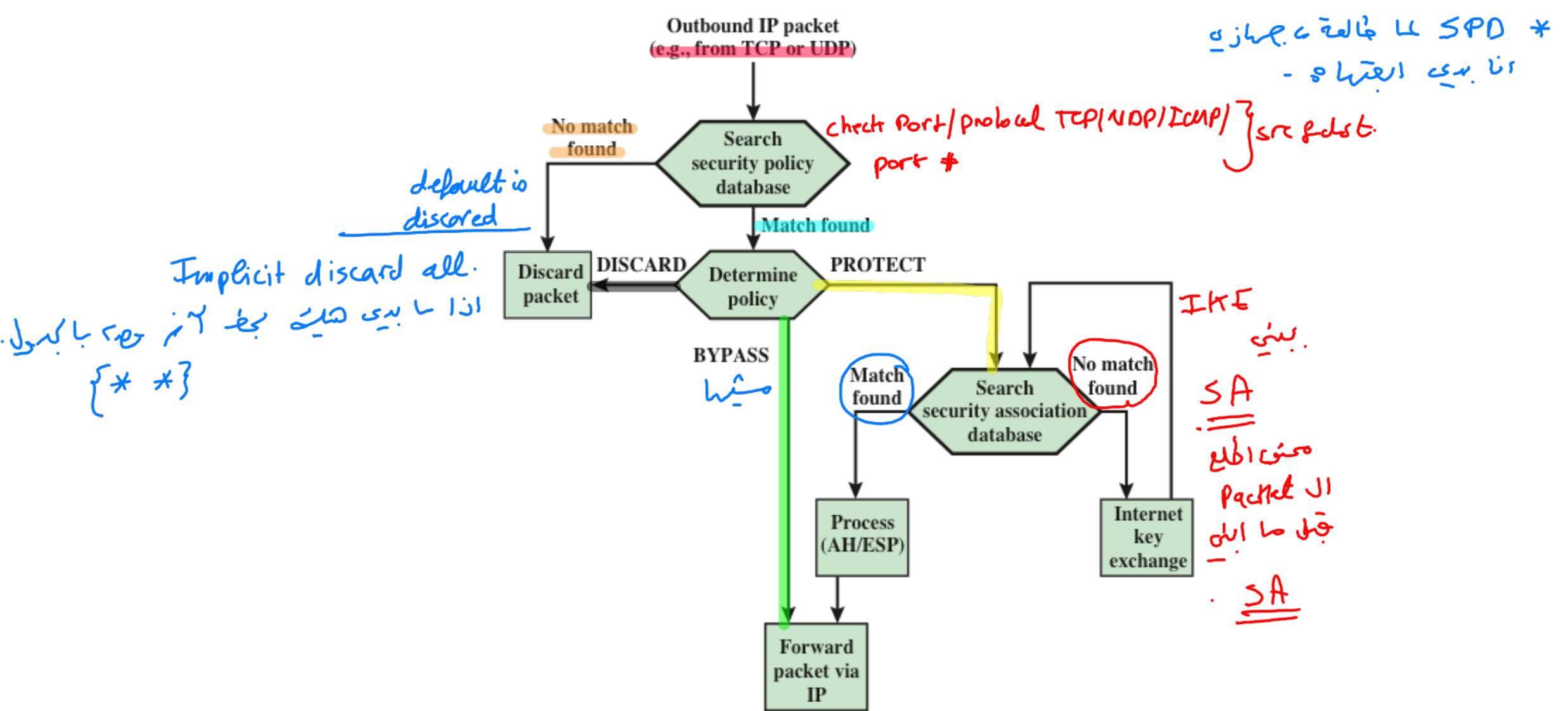


Figure 20.3 Processing Model for Outbound Packets

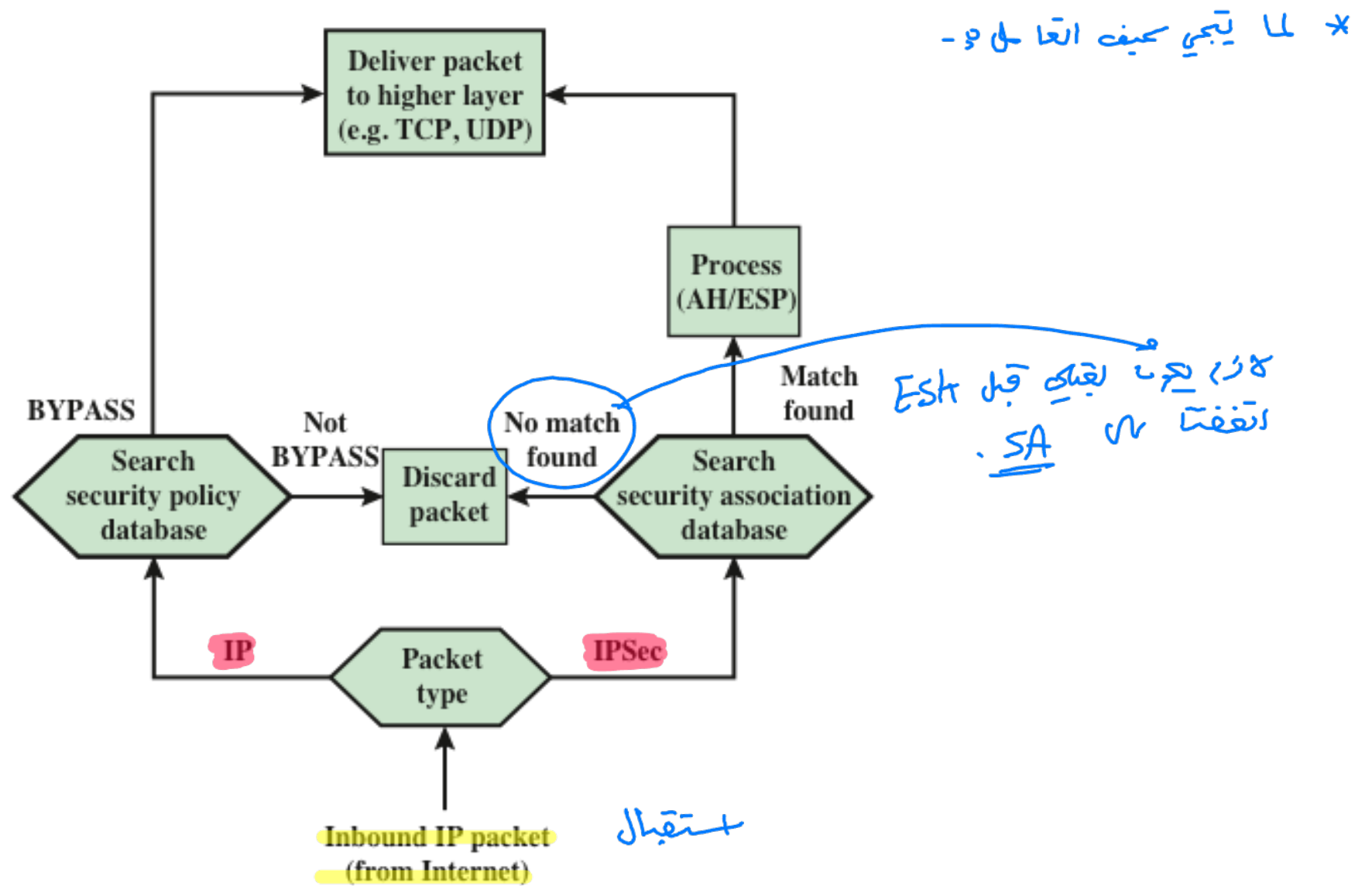


Figure 20.4 Processing Model for Inbound Packets

* Window size with sequence numbers -

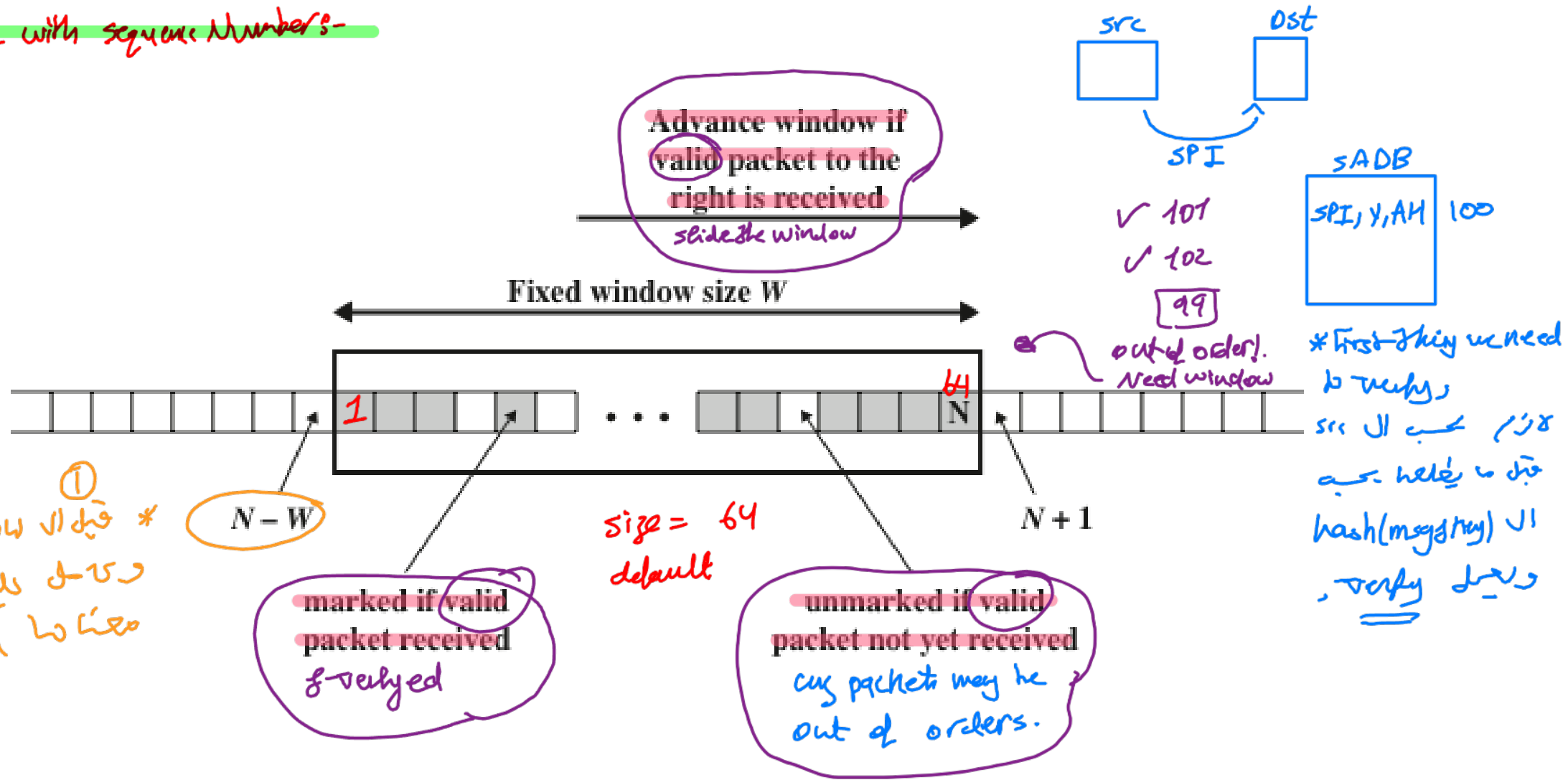
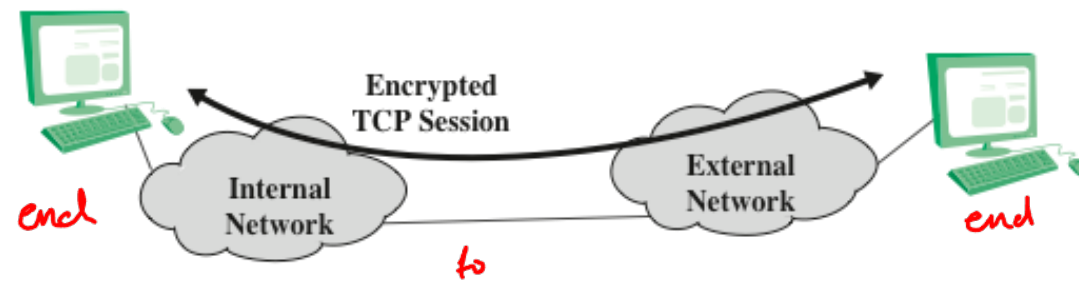


Figure 20.6 Anti-Replay Mechanism



(a) Transport-level security

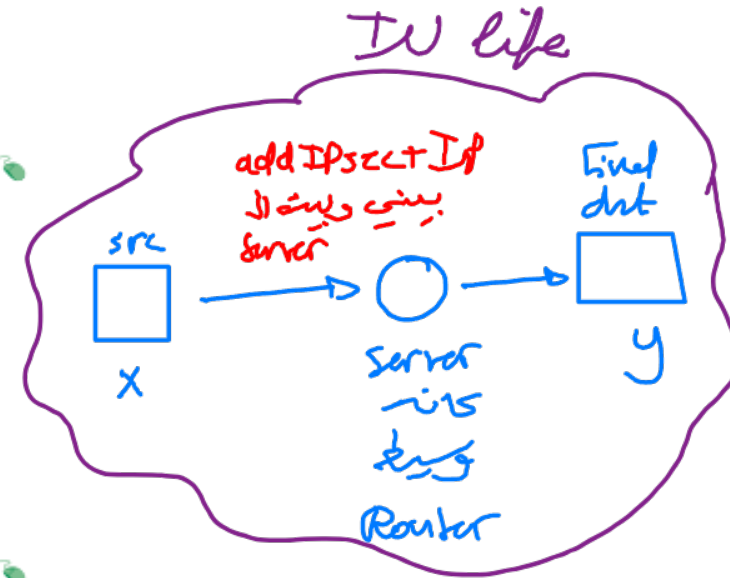
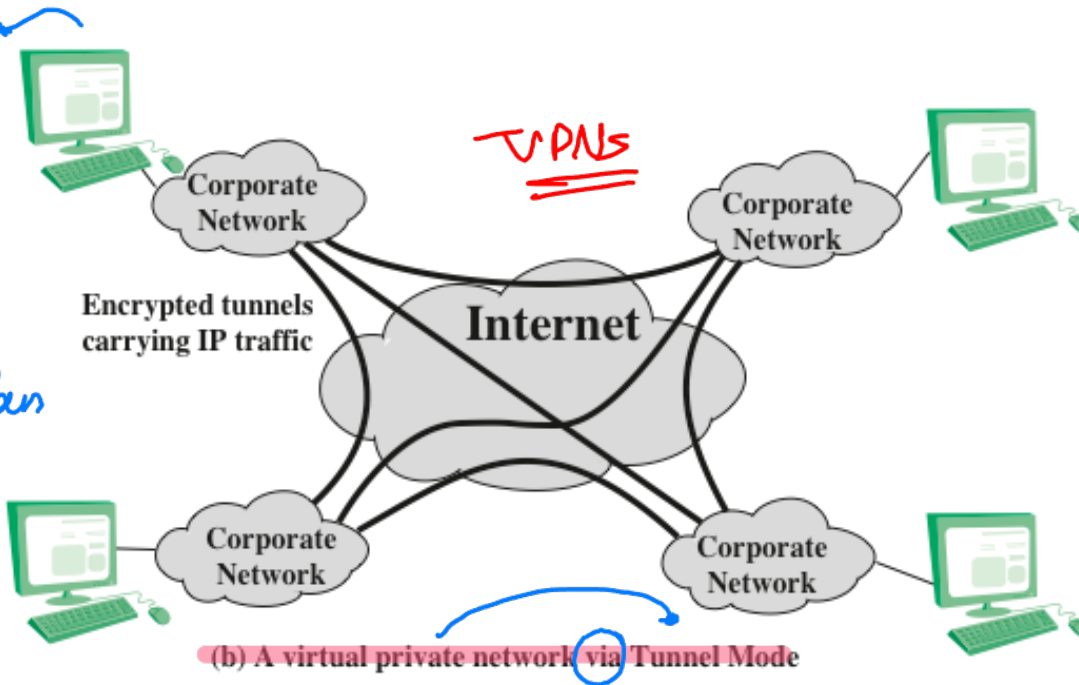


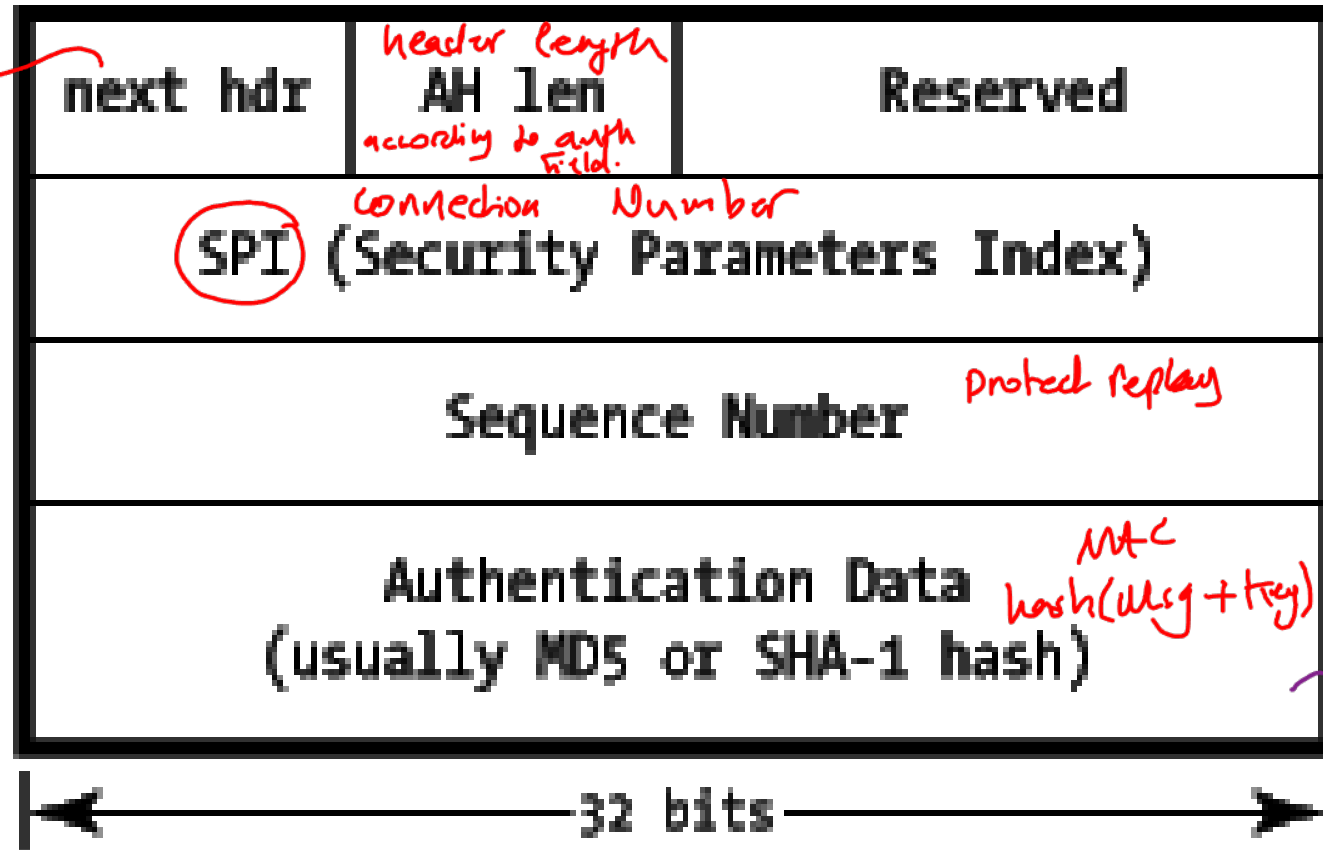
Figure 20.7 Transport-Mode vs. Tunnel-Mode Encryption

more than one campus
 بدم الحوال traffic خف
 البتكا و بينم هتو يتوا
 الا خف خف كاين هولت ل
 single cam
 any talk between them encrypted
 any traffic to other place discarded
 it.

AH HEADER :- either AH or ESP.

~~IPSec AH Header~~

all protocols on all layers
جس header موجود!
فیلڈ بیسے کی ال fields



برف کن ہریتہ فکرت
بال
SAP

AH Transport Mode

Original IPv4 Datagram

ver	hlen	TOS	pkt len	
ID			flgs	frag offset
TTL	proto=TCP	header cksum		
src IP address				
dst IP address				

TCP header (proto = 6)

TCP payload

Protected by
AH Auth Data

New IPv4 Datagram

ver	hlen	TOS		pkt len + AH size	
ID			flags	frag offset	
TTL	proto=AH		header cksum		
src IP address					
dst IP address					
next=TCP		AH len	Reserved		
SPI (Security Parameters Index)					
Sequence Number					
Authentication Data (usually MD5 or SHA-1 hash)					
TCP header (proto = 6)					
TCP payload					

Northstar 1140
Ilse go

IPsec header

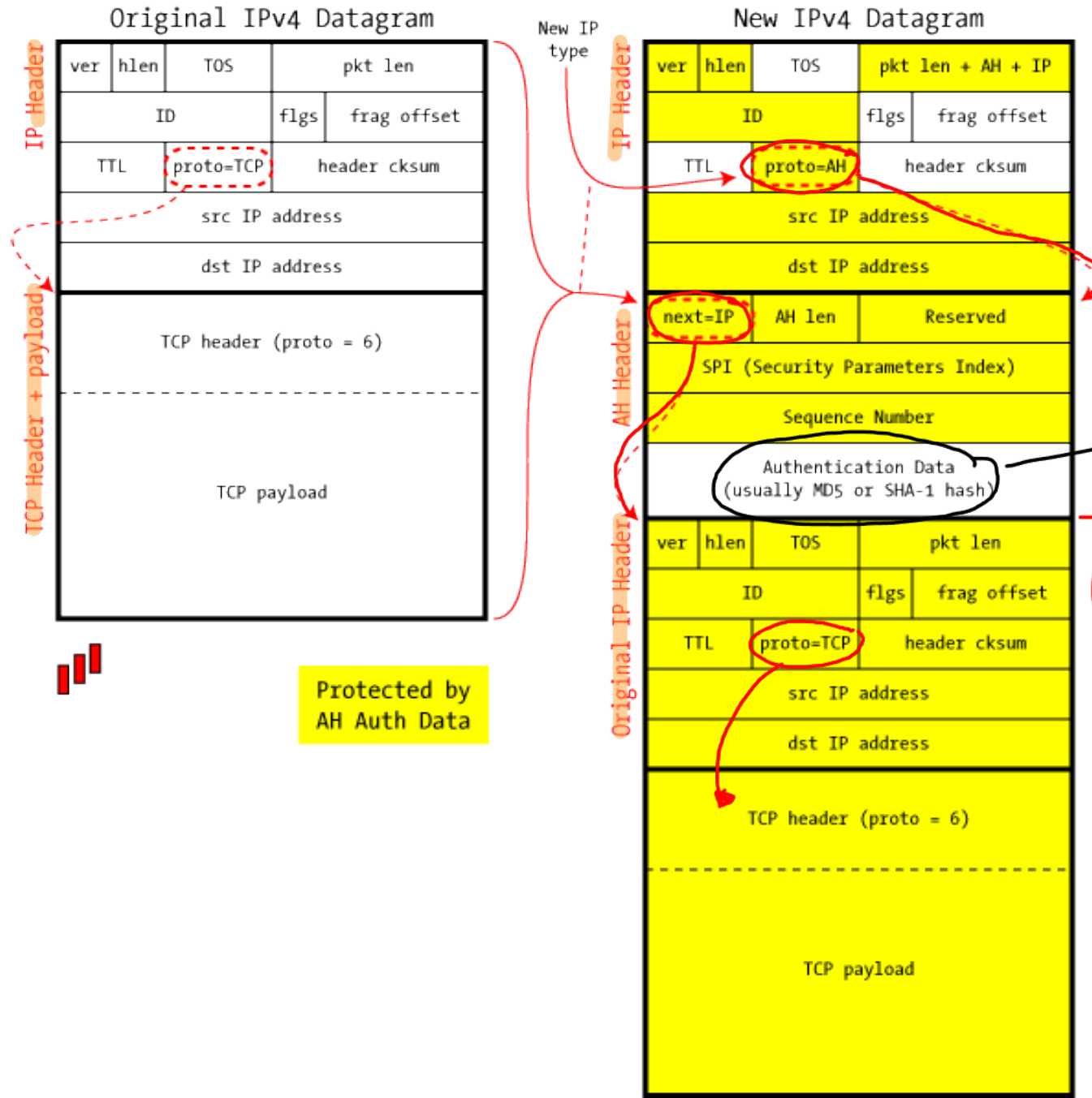
سَوَّبَ لِحَالِ اللَّهِ بِالْجَزْ
مَعْدَا الْإِبْرِيَّةِ لَا تَوْ
بَتَغْدِرُ along the path

Next header
بَاقِي

TCP

APP

IPSec in AH Tunnel Mode



بجای کل الدافل لازم
بغل بآلنق فیرا، صا
یاغوبالو فوق .
بجای ال Packet و صا
+ parts from external
Ip header.

خفیل ز صا

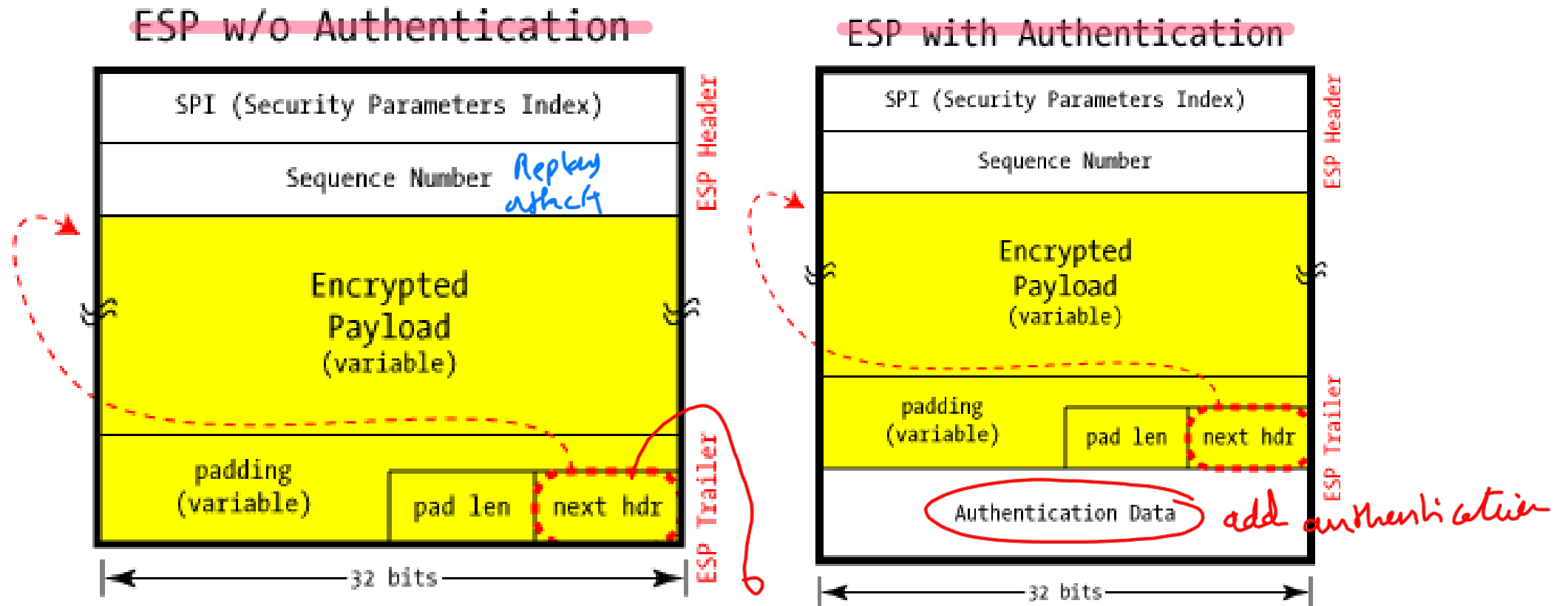
ENCAPSULATING SECURITY PAYLOAD (ESP)

- Used to encrypt the payload data
- Optional auth is present
 - ~~ICV is computed after the encryption is performed~~ ^{default first} on ESP
 - Because the ICV is not protected by encryption, a keyed integrity algorithm must be employed to compute the ICV

authenticate you computed for encrypted msg with the original msg.

→ $\text{hash}(\text{encrypted msg} + \text{key})$

ESP HEADER



when you decrypt the data.

ESP PADDING

1) I need it cuz data is encrypted & encrypted data may be need block encryption technique.
padding ادا ی سته اقل بفر اعل padding 128/256.

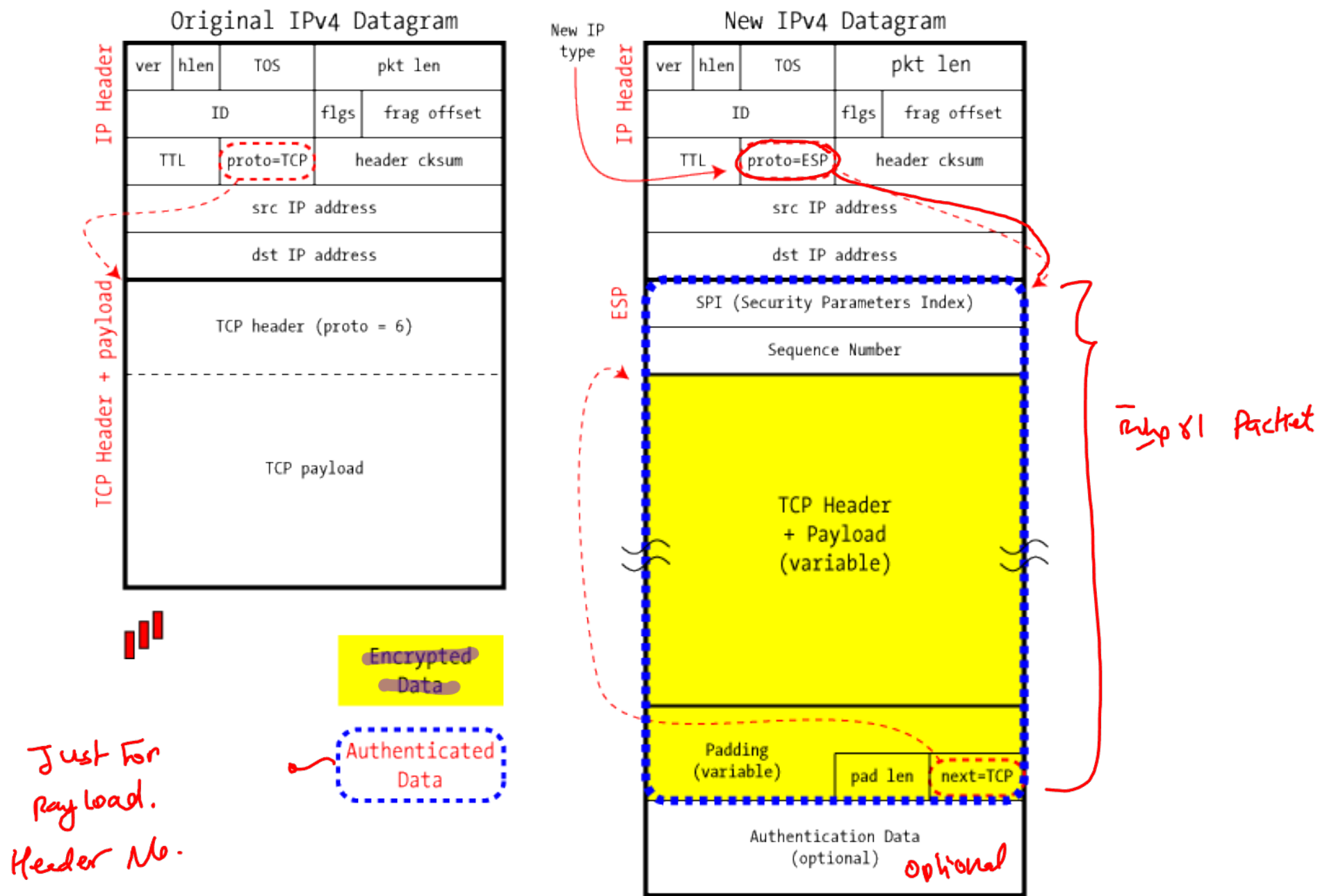
■ The Padding field serves several purposes:

- 1) If an encryption algorithm requires the plaintext to be a multiple of some number of bytes.
- 2) Used to assure alignment of Pad Length and Next Header fields
- 3) Additional padding may be added to provide partial traffic-flow confidentiality by concealing the actual length of the payload

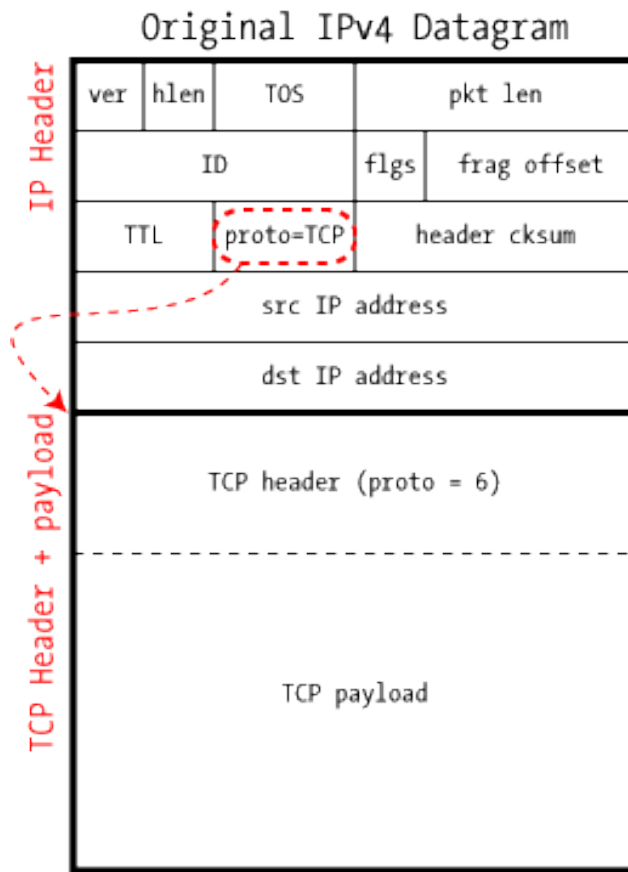
2) word multiple of 32-bits
padding خازا سته لقل لعل padding
3) do padding for personal use
ex:- packet ال padding

* بعض ادا padding ال original data
padding خازا ال padding

IPSec in ESP Transport Mode

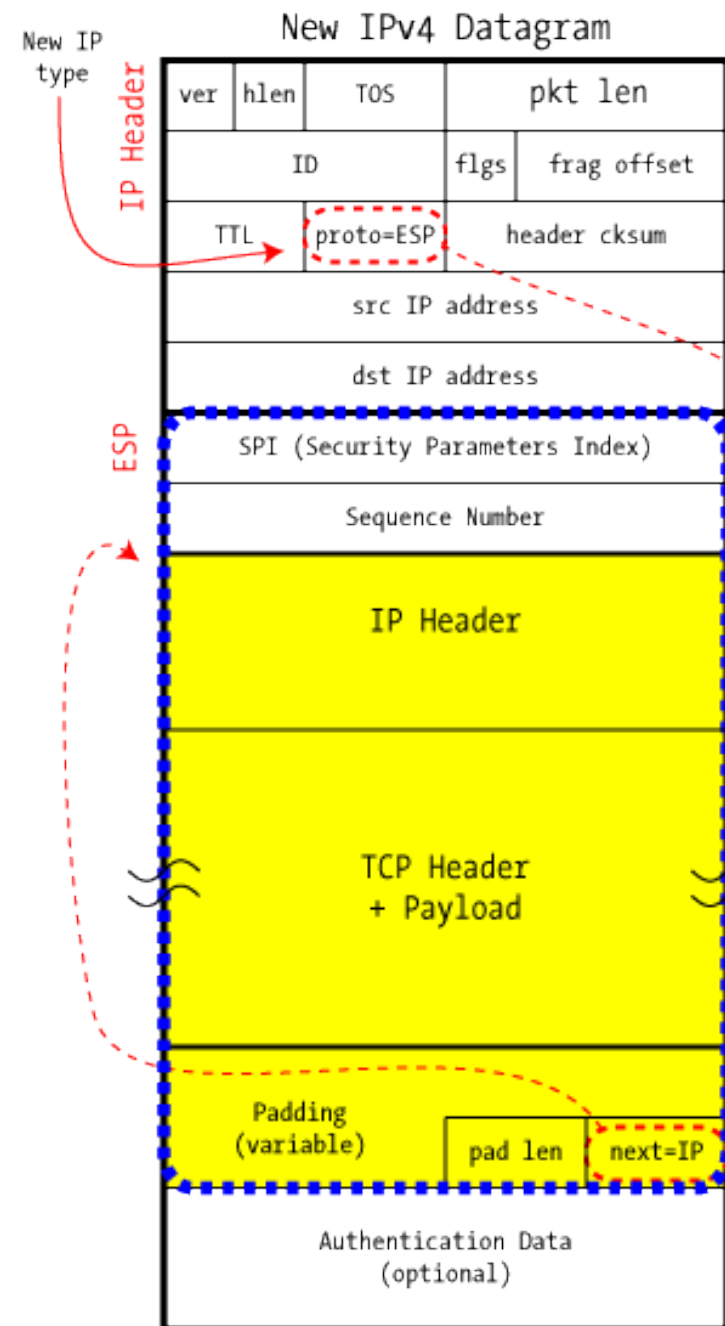


IPSec in ESP Tunnel Mode



Encrypted Data

Authenticated Data



Original Packet authentication header is added
ESP tunnel mode.

زیر سر و سر
packet
encryption header
SPI و
sequence number

COMBINING SECURITY ASSOCIATIONS

- An individual SA can implement either the AH or ESP protocol but not both
- Security association bundle
 - A sequence of SAs through which traffic must be processed to provide a desired set of IPsec services
- May be combined into bundles in two ways:
 1. Transport adjacency: Refers to applying more than one security protocol to the same IP packet without invoking tunneling. This approach allows for only one level of combination
 2. Iterated tunneling: Refers to the application of multiple layers of security protocols effected through IP tunneling. This approach allows for multiple levels of nesting

ESP WITH AUTHENTICATION OPTION

- A user applies first encryption to the data to be protected and then appends the authentication data field.
- Transport mode ESP
 - Authentication and encryption apply to the IP payload delivered to the host, but the IP header is not protected
- Tunnel mode ESP
 - Authentication applies to the entire IP packet delivered to the outer IP destination address and authentication is performed at that destination
 - The entire inner IP packet is protected by the encryption algorithm for delivery to the inner IP destination
- For both cases authentication applies to the ciphertext rather than the plaintext

TRANSPORT ADJACENCY

- A way to apply authentication after encryption in ESP is to use two bundled **transport** SAs, with the inner being an ESP SA and the outer being an AH SA
 - In this case ESP is used without its authentication option
 - Encryption is applied to the IP payload
 - AH is then applied in transport mode
 - Advantage of this approach is that the authentication covers more fields
 - Disadvantage is the overhead of two SAs versus one SA

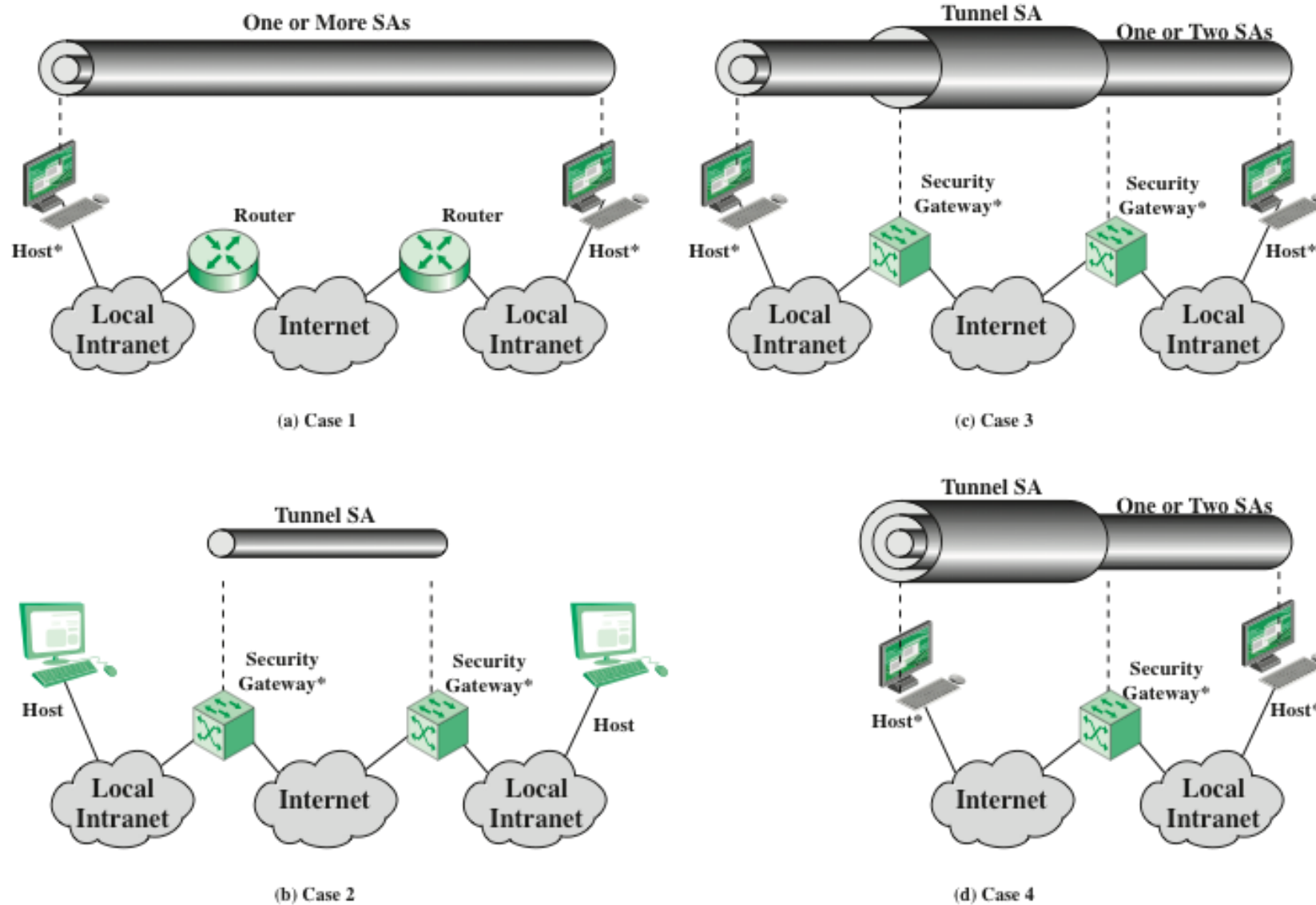
TRANSPORT-TUNNEL BUNDLE

The use of authentication prior to encryption might be preferable for several reasons:

- It is impossible for anyone to intercept the message and alter the authentication data without detection
- It may be desirable to store the authentication information with the message at the destination for later reference

One approach is to use a bundle consisting of an **inner AH transport SA** and an **outer ESP tunnel SA**

- Authentication is applied to the IP payload plus the IP header
- The resulting IP packet is then processed in tunnel mode by ESP
 - The result is that the entire authenticated inner packet is encrypted and a new outer IP header is added



* = implements IPsec

Figure 20.10 Basic Combinations of Security Associations

INTERNET KEY EXCHANGE

- The key management portion of IPsec involves the determination and distribution of secret keys
 - A typical requirement is four keys for communication between two applications: Transmit and receive pairs for both integrity and confidentiality
- The IPsec Architecture document mandates support for two types of key management:
 1. Manual
 - A system administrator manually configures each system with its own keys and with the keys of other communicating systems
 - This is practical for small, relatively static environments
 2. Automated
 - Enables the on-demand creation of keys for SAs and facilitates the use of keys in a large distributed system with an evolving configuration

ISAKMP/OAKLEY

- The default automated key management protocol of IPsec consists of:
 - Oakley Key Determination Protocol
 - A key exchange protocol based on the Diffie-Hellman algorithm but providing added security
 - Generic in that it does not dictate specific formats
 - Internet Security Association and Key Management Protocol (ISAKMP)
 - Provides a framework for Internet key management and provides the specific protocol support, including formats, for negotiation of security attributes
 - Consists of a set of message types that enable the use of a variety of key exchange algorithms

SUMMARY

IP security overview

- Applications of IPsec
- Benefits of IPsec
- Routing applications
- IPsec documents
- IPsec services
- Transport and tunnel modes

IP security policy

- Security associations
- Security association database
- Security policy database
- IP traffic processing
- Cryptographic suites



- Encapsulating security payload
 - ESP format
 - Encryption and authentication algorithms
 - Padding anti-replay service
 - Transport and tunnel modes
- Combining security associations
 - Authentication plus confidentiality
 - Basic combinations of security associations
- Internet key exchange
 - Key determination protocol
 - Header and payload formats